

TEST & MEASUREMENT

Fuelling Technological Advances

Imbibing progressive practices ranging from multicore (parallel) processors and software-defined instrumentation to wireless standards, the field of test and measurement continues to enable new technologies



Software-defined instrumentation (Courtesy: National Instruments)

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The use and development of technology requires test and measurement instruments. From research and development of new technologies to ensuring quality in the manufacturing of various products, these instruments are essential. For example, when a space shuttle is launched, many different instruments are used to conduct thousands of electrical, temperature, chemical and other measurements to monitor the status

of the craft and its crew. These instruments are essential to our high-tech economies," says Dr Anita Agrawal, assistant professor in department of electrical and electronics engineering, BITS, Pilani-K.K. Birla Goa Campus, opening our discussion on developments and trends in the test and measurement (T&M) space.

The goal of T&M has often been misunderstood as mere fault-finding. Of course, the detection of faults during manufacture has been a key task handled by T&M instruments; and has been instrumental in achieving high

quality-control levels in all technological arenas. However, T&M plays a larger role in monitoring and maintaining various critical technical scenarios, and in the research, validation and improvement of new technologies.

Further, the relationship between T&M and other technological areas has always been intertwined and symbiotic. While T&M helps develop new technologies, it also imbibes the best of technologies to advance its own capabilities further. Progress in networking and telecommunications, for example, has been possible only because of capable T&M equipment that can accurately measure and monitor various critical technical and quality-of-service (QoS) parameters on a real-time basis. At the same time, fast wireless and Ethernet technologies have helped make T&M instruments more communicative and capable of connecting to larger enterprise systems via fast networks. It is this symbiotic relationship that drives a lot of improvements and innovations in the field of test and measurement.

Inspiration to improve— from technology and technologists

The innovations in T&M cater to both the needs of new technologies as well as the demands of technologists and manufacturers.

At one time, strict quality regulations and the need to find even the smallest of faults were the key innovation drivers in the T&M industry. But now high standards have been set and